

04.01 Radical Expressions

Concept summary and key formulas

Key ideas

A radical is simplified by factoring out perfect powers.

Like radicals can be combined only after their radicands match.

For products and quotients, use the product and quotient properties, then simplify.

Key formulas

$$\begin{aligned}\sqrt{ab} &= \sqrt{a}\sqrt{b} \\ \sqrt{\frac{a}{b}} &= \frac{\sqrt{a}}{\sqrt{b}} \\ \sqrt{a^2} &= |a|\end{aligned}$$